

Identifying EV Bad Behaviour: The Parking Lot Usage Efficiency (PLUE) Metric

Stephen Makonin*, Patrick Palmer†, Fred Popowich‡, and David Agosti§,

*School of Engineering Science, Simon Fraser University, Burnaby, Canada

†School of Sustainable Energy Engineering, Simon Fraser University, Surrey, Canada

‡SFU's Big Data Hub, Simon Fraser University, Burnaby, Canada

§Parking & Sustainable Mobility Services, Simon Fraser University, Burnaby, Canada

Email: smakonin@sfu.ca, patrick_palmer@sfu.ca, popowich@sfu.ca, david_agosti_2@sfu.ca

Abstract—We observe that EV charging stalls are often underutilized when drivers leave vehicles plugged in after charging ends, undermining station throughput, frustrating users, and complicating the business case for smart-parking investments. To address this gap, we introduce the parking lot usage efficiency metric (PLUE), defined as the ratio of active charging time to total plug-in time. PLUE quantifies post-charge occupancy on a continuous scale from zero (fully idle) to one (fully efficient), enabling objective classification of user behaviour and identification of inefficiencies. We demonstrate PLUE's utility by applying it to a real-world multi-year campus charging dataset, revealing persistent idle intervals across multiple stations despite generally consistent charging demand. Parking Operators can then use the PLUE score together with operational KPIs like length-of-stay to better inform decision making such as matching infrastructure power levels to time parked, establishing overstay fines, and various other behaviour and revenue metrics.

Index Terms—Electric Vehicles, Parking Lot, Usage Efficiency, Charging Behaviour, Range Anxiety, L2 Chargers

I. INTRODUCTION

Electric vehicle (EV) parking and charging etiquette is essential for ensuring that shared charging infrastructure serves all users efficiently and harmoniously. Common breaches such as vehicles occupying charging spots without actively charging, failure to move once charging is complete, or blocking access with non-EV vehicles can frustrate drivers and reduce stall availability [1]–[3]. Industry guides stress the importance of promptly vacating stalls once charging ends, parking within designated bay lines, and respecting queuing norms to foster positive EV-driver etiquette [2], [3]. Beyond courtesy, these behaviours directly affect utilization rates: unoccupied chargers can sit idle for extended periods, undermining station throughput and exacerbating peak-demand pressures on the grid [4]. Additionally, from a business perspective, parking operator courtesy needs to look at customer satisfaction as it is an important metric for return business — a qualitative measure hard to capture within a quantitative dataset.

To address these challenges, many operators are exploring smart-parking technologies such as real-time occupancy

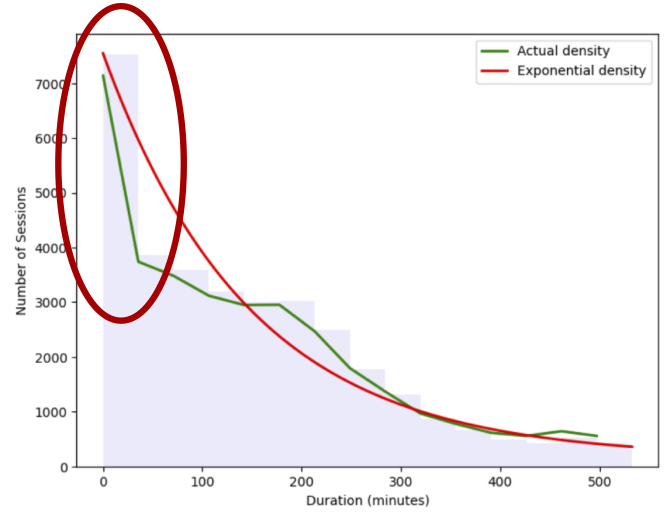


Fig. 1. Figure originally from [5]. Identifying range anxiety (circled in red). Curiosity: does Range anxiety also leave to bad parking Efficiency in the case of other sessions, with drivers leaving cars after they are fully charged?

sensors, mobile alerts, dynamic pricing, and automated enforcement with the promise of boosting return on investment (ROI) by up to 30–37% over several years [6]. However, the high upfront costs and often lengthy payback periods make it difficult for stakeholders to justify such capital expenditures without clear, quantifiable evidence of misbehaviour impacts [4], [6]. Meanwhile, behavioural factors like range anxiety (clearly identified in Fig. 1) where drivers indulge in large numbers of short sessions unnecessarily. This can further degrade parking-lot efficiency with delayed return to their vehicles after charging completes, and remains poorly understood in quantitative terms [4]. This paper introduces a *metric* to objectively measure “BAD” charging-stall occupancy, providing the empirical foundation needed to calculate ROI and guide investment decisions.

A. L2 Charging and the Overstay Business Case

The motivation here focuses on L2 charging stations as opposed to L3 or DCFC charging stations, which would have different charging habits, charging tools/hardware, and charging environments. L2 charging stations are increasing installed

Funded by Natural Resources Canada (NRCan) under project Evid-3001 named *Tap&Go Electric Vehicle Charging Infrastructure Demonstration* and Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant RGPIN-2018-06192.

in many parking lots (e.g., shopping malls and community centres). For parking lots with low-efficiency, parking lot managers may wish to implement/impose an overstay penalty. However, if there is a parking fee that “rounds” up at the hour, it may be desirable to focus on customer satisfaction versus revenue generation. As will be described in this paper, the PLUE metric can be used for both parking lot usage efficiency and the specific efficiency of the charger/charging space — macro-level and micro-level analytic can be achieved. Being able to measure efficiency will allow parking lot managers to implement the policies that make sense for their parking lot. An additional part of the PLUE metric can help to determine if a parking site is at, above, or below charging capacity.

II. BACKGROUND

Electric vehicle charging behaviour has been the focus of extensive empirical and modelling studies, leveraging rich datasets to inform both algorithm development and infrastructure planning. Lee *et al.* introduce ACN-Data [7], a publicly available, dynamically updated dataset comprising over 30,000 workplace charging sessions collected at two California sites managed by PowerFlex Systems. In their work, they employ Gaussian mixture models to uncover multi-modal arrival and departure patterns, propose methods for optimally sizing on-site solar generation to support adaptive charging, and demonstrate how workplace charging control can smooth the grid’s “duck curve” of net demand. While ACN-Data inherently records the full session duration, from plug-in to unplug, Lee *et al.* focus their analysis on charging load profiles and algorithm validation, without explicitly isolating or examining the post-charge idle interval during which vehicles continue to occupy stalls.

Building on the proliferation of such datasets, Shahriar *et al.* [8] offer a comprehensive review of machine learning approaches applied to EV charging behaviour, covering supervised and unsupervised methods as well as deep neural networks. They categorize the main tasks—session clustering, user typology classification, and regression of session metrics such as duration and energy consumed and discuss key model families including k-nearest neighbours, support vector machines, and deep learning architectures. The authors also highlight future research directions like integrating high-dimensional features (e.g., traffic, weather) and applying reinforcement learning to scheduling problems. Despite this breadth, their survey concentrates on predicting charging demand and duration, with no treatment of the subsequent parking occupancy that affects space availability.

Prasad *et al.* (2023) [9] take a data-driven next step by enhancing the ACN-Data dataset with traffic counts, charging current measurements, and weather variables, then training ensemble machine learning models to predict both entire session duration (mean absolute error of 56 minutes) and active charging duration (MAE of 43.2 minutes), as well as energy consumption (best MAE $\approx$ 1.36 kWh). By distinguishing total plug-in time from actual charging time, they indirectly reveal the presence of idle intervals; however, their analysis stops

short of characterizing this idle period as a metric of parking stall occupancy or turnover.

Collectively, these studies advance our understanding of EV charging patterns and predictive modelling, yet none directly quantify the post-charge dwell time that impacts parking lot usage efficiency. This gap motivates the development of the PLUE metric, which explicitly quantifies how charging and post-charging parking behaviours together impact parking lot efficiency.

TABLE I
BEHAVIOUR LABELS FOR EFFICIENCY, OVERSTAY AND CAPACITY

Behaviour	Efficiency Range	Overstay Range	Capacity Range
GOOD	1.00 – 0.80	0.00 – 0.20	—
NORMAL	0.79 – 0.60	0.21 – 0.40	0.00 – 0.89
BAD	0.59 – 0.00	0.41 – 1.00	0.90 – 1.00

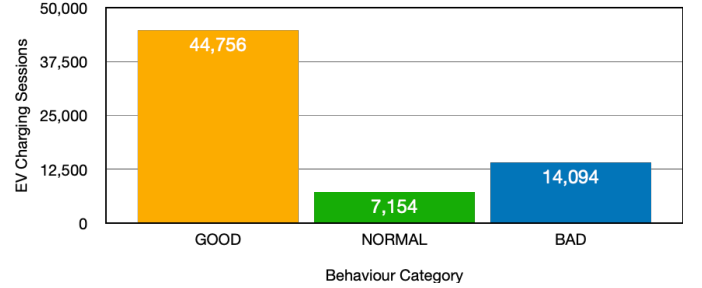


Fig. 2. How all sessions (y-axis) group in each efficiency category. Category bars are sort in descending order.

III. METHODOLOGY

The *parking lot usage efficiency* (PLUE) metric combines charging and parking behaviours to assess how effectively charging infrastructure and associated capacity of each parking spaces is utilized. The PLUE metric can be simplify defined as:

$$\text{PLUE}(l) = \text{Efficiency}(l) @ \text{Capacity}(l) \quad (1)$$

where l represents a particular EV charging station y in a parking spot or all EV charging stations in a particular parking lot z . An EV charger score y example would look like:

$$\text{PLUE}(\text{Carger0001}) = 0.2 @ 95\% \text{ capacity.} \quad (2)$$

This PLUE means that on average there is a very BAD Efficiency Score of 0.2 with a very BAD/high Capacity (utilization) Score of 95% for this parking lot. This means the overstay of EV vehicles is very problematic and with a high capacity adding an additional EV charger should be a high priority task. For a parking lot score z example:

$$\text{PLUE}(\text{P3Parkade}) = 0.5 @ 60\% \text{ capacity.} \quad (3)$$

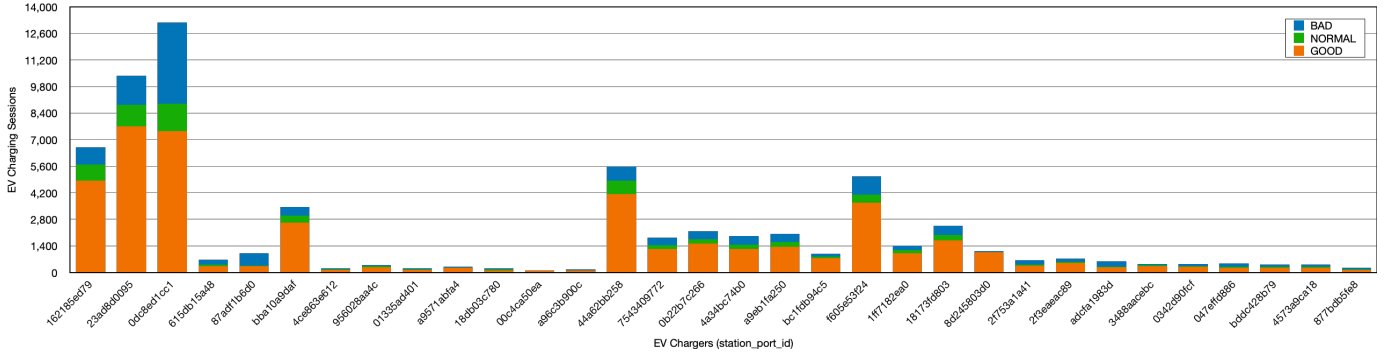


Fig. 3. The Overstay Score for all EV charging stations for the entire dataset (since 2019). Taller bars mean that the particular EV charging station has more charging sessions either for more usage or was installed earlier than the others. This essentially a breakout of the results shown in Fig. 2.

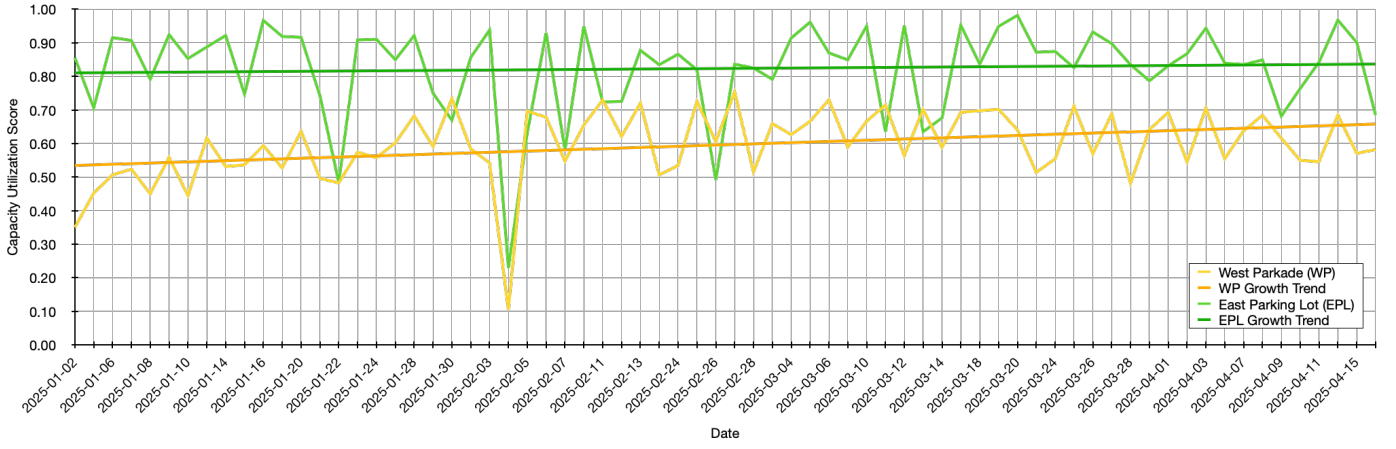


Fig. 4. Capacity utilization and growth analysis for 15 Chargers in both West Parkade (WP) and East Parking Lot (EPL) for time range (working days) Jan 1 to April 16, 2025. The WP capacity growth tend is $y = 0.0018x + 0.53$, $R^2 = 0.1212$ and for EPL the growth tend is $y = 0.0004x + 0.8102$, $R^2 = 0.0033$. The noticeable dip in capacity on Feb 4, 2025 was due to the SFU Burnaby Mountain campus being closed due to a large snowfall over night).

This PLUE means that on average there is a BAD Efficiency Score of 0.5 with a NORMAL Capacity (utilization) Score of 60% for this parking lot. This means the overstay of EV vehicles is problematic but might not require urgent attention since the availability of free EV parking spots is high due to a low capacity utilization.

How efficiently a parking spot with an EV charging station is used can be viewed as how close the EV vehicle is to unplugging after charging has completed. The efficiency of an EV charging session is inversely related to the length of time a vehicle remains parked after charging completes: the longer the post-charge idle interval, the lower the utilization efficiency. This can be represented by a simple normalized value defined as:

$$\text{Efficiency}(z) = \frac{\sum_y^Y \sum_x^X \Delta \text{charging}_t(y[x])}{\sum_y^Y \sum_x^X \Delta \text{plugged-in}_t(y[x])} \quad (4)$$

where z is a given parking lot, Y are the charging stations located in parking lot z , X are the charging sessions for charging station y . Further, x is a given session, $\Delta \text{charging}_t(y[x])$ is the time spent charging at EV charging station y session

x , $\Delta \text{plugged-in}_t(y[x])$ is the time spent plugged-in at EV charging station y session x , and Δ represents the amount of time between start and end timestamps. Note that the summation over Y can be removed for calculation of just a particular EV parking spot.

The Overstay Score is basically the opposite of the Efficiency Score (i.e., or negative representation) and defined as:

$$\text{Overstay}(z) = 1 - \text{Efficiency}(z). \quad (5)$$

Another aspect to the PLUE metric and has a impact on the importance of planning how to deal with a bad overstay score is understanding the capacity utilization each parking spot and indeed the whole parking lot. We define this as the Capacity Score (i.e., capacity utilization percentage):

$$\text{Capacity}(z) = \frac{\text{time plugged-in}}{\text{total available time}}. \quad (6)$$

where the capacity utilization percentage is simply $\text{Capacity}(z) \times 100\%$. An average is taken when calculating the capacity utilization spanning multiple available times; e.g., peak parking hour for the previous month.

As we calculate the PLUE score, where higher scores indicate efficient use given the capacity. Once every stall's PLUE score is calculated, we apply predefined efficiency bands to classify user behaviour. Table I lists the ranges we deemed reasonable—based on typical parking-lot usage patterns and ROI objectives. However, these thresholds can be tuned to reflect different lot configurations or operational goals. These categories will then guide our analysis in the next section.

IV. DATA ANALYSIS

For the analysis the SFU-EVP dataset [5] (April 17, 2025 update) was used. SFU-EVP encompasses 66,009 EV charging sessions recorded between November 6, 2019, and mid-day April 17, 2025, using ChargePoint infrastructure on the Simon Fraser University campus. Over that period, 3,539 distinct users logged sessions, consuming anywhere from 0.00 kWh up to a peak of 137 kWh per session. Most charging events (43,275) fall below 14.5 kWh, 6,581 sessions have 0 kWh, with an average draw of 13.8 kWh. This is consistent with the energy needs of daily commuters topping up their batteries during work hours.

When examined on a weekly basis, session energy remains remarkably stable, with the average of 10.28 kWh translating to roughly an 80 km range per charge—slightly above typical daily driving distances. The bulk of sessions are modest in kWh delivered, reflecting “topping off” behaviour rather than deep discharges. There is no evidence of outlier driving patterns among SFU-EVP users; charging habits align closely with standard daily usage, making the dataset well suited for developing mathematical models of campus EV parking and charging efficiency.

When calculating the PLUE score for an entire parking lot, of which SFU has several (i.e., all chargers in SFU-EVP), we find that the parking lot has a normal efficiency, scoring 0.783; however, there are an overwhelming number of sessions that are in the GOOD efficiency category, Fig. 2, which depicts the amount of sessions that fall in each category. This might indicate that there is a wide range of user types, which we anticipate, such as shoppers, students, staff and residents.

Looking at each charging station, we found that in 2025 2 of the 15 charging stations are in the GOOD efficiency category with NORMAL capacity utilization. A more detailed look at the results for each charging station can be found in Fig. 3, where their location is indicated and their ratios of good, bad and normal sessions are shown. We show the PLUE metric results in Fig. 5 as a plane to better interpret the metric results.

For capacity utilization, (see Fig. 4) we note that the one EV charger in the East Parking Lot (EPL) has a slow trending growth in capacity, above 80%, which make the lot a very good candidate for adding more EV chargers. On the other hand, West Parkade (WP), which already has 14 EV chargers, has a faster trending growth in capacity and is forecasted to surpass the 90% capacity level in about 205 days from April

16 (the last day used in our analysis). November 8 will be close to the day where capacity is steadily above 90%.

Range anxiety does seem to have a negative effect on the efficient use of some parking stalls. In contrast, others in similar locations have little problem, attracting those with mostly good behaviour. It is likely that good behaviour is encouraged, when it is easily possible for the driver to relocate the vehicle to non-EV spaces. This could be determined by the general availability of the parking or the proximity of the parking to the driver while parked.

Although the PLUE determination according to our methodology may indicate bad behaviour from an efficiency and cost effectiveness point of view, especially when the intent is to deliver energy, range anxiety is a valid concern for the user and their satisfaction. Clearly, applying any penalties for bad PLUE score behaviour would need careful consideration on the basis that overcoming range anxiety for EV users is part of increasing the wider acceptability of EVs.

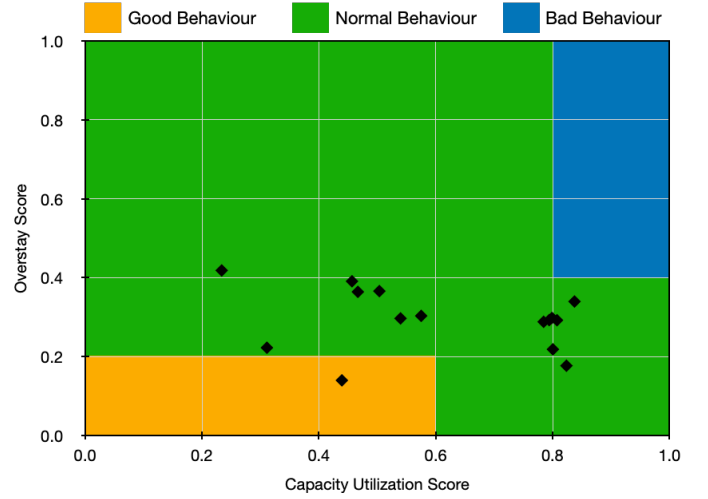


Fig. 5. The PLUE Plane for 15 Chargers in both West Parkade and East Parking Lot for time range (working days) Jan 1 to April 16, 2025. For the plane Overstay Score is used rather than Efficiency Score so that higher or “bad” scores are further from the x and y axis (i.e., origin of the plane).

A. Metric Flexibility

The PLUE score is a 24/7 metric, so different time periods in mixed-use parking lots would have very different usage patterns and revenue objectives. For example, people who park overnight can be measured separately and treated differently from those who park during regular business hours, which can result in very different good or bad PLUE scores for each time/use period.

The GOOD efficiency range of > 0.8 is the right threshold we use for our data analysis, but it can be changed based on the goals and needs of different parking lot managers. The parking option goals and needs at SFU can be very different than those of a shopping mall. For example, at a shopping mall (2 hours of parking), a good rating equals 40 minutes idle or less, while parking at a worksite (8 hours of parking) requires a good rating of 90 minutes idle or less.

V. CONCLUSIONS

The *parking lot usage efficiency* (PLUE) metric offers a focused lens on EV drivers' post-charge parking behaviours, quantifying how long vehicles occupy spaces after charging completes with respect to the capacity utilization of that spot. Applied at the level of individual charging stations or across an entire parking facility, PLUE highlights instances of inefficient use along with capacity utilization issues and enables targeted interventions. However, PLUE represents just one component of a comprehensive strategy to curb undesirable parking practices. Effective solutions will blend driver education on proper charging etiquette, infrastructural design improvements to encourage turnover, and technological innovations such as automated session reminders or dynamic pricing.

A. Future Work

Looking ahead, expanding the suite of behavioural metrics—particularly those that capture demand fluctuations and queuing dynamics—will deepen our understanding of EV user patterns and support more nuanced management of charging ecosystems. We also want to measure and analyze the effects of mixed-use sites where certain stakeholders may contribute differently to PLUE due to their different behaviours (e.g., at a

shopping mall – shoppers vs. staff) and may therefore respond better/worse to different interventions.

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